



COMSATS University Islamabad, Lahore Campus

Department of Computer Engineering

Quiz # 1

Course Title:	VLSI Design	Course Code:	CPE434	Credit Hours:	4(3,1)
Course Instructor:	Dr. Muhammad Naeem Awais	Programme Name:	BS Computer Engineering		
Semester:		Batch:		Date:	5 th March 2026
Time Allowed:	10 Minutes		Maximum Marks:	5	
Student's Name			Registration. No.:		

Question # 1:

(PLO3-CLO3-C5)

Design a CMOS circuit of the following function using an implementation comprised only of CMOS NOR gates. Assume true and complemented inputs are available.

$$Y = S_2'S_1'S_0'D_0 + S_2'S_1'S_0'D_1 + S_2'S_1S_0'D_2 + S_2'S_1S_0D_3 + S_2S_1'S_0'D_4 + S_2S_1'S_0D_5 + S_2S_1S_0'D_6 + S_2S_1S_0D_7$$



